

Name: C.Haritha

Reg No:- 23BEC7347

Spring REST Using Spring Boot 3

1) Spring web project using maven

SpringLearnApplication.java

```
package com.cognizant.springlearn;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication public class
SpringLearnApplication {    public static
void main(String[] args) {
    System.out.println("Application Starting...");
    SpringApplication.run(
        SpringLearnApplication.class,
args);
    System.out.println("Application Started...");
    }
}
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.0</version>
    <relativePath/>
  </parent>
  <groupId>com.cognizant</groupId>
  <artifactId>spring-learn</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>spring-learn</name>
  <properties>
    <java.version>17</java.version>
  </properties>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-devtools</artifactId>
```

```
        <scope>runtime</scope>
        <optional>true</optional>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>
</project>
```

application.properties

```
spring.application.name=spring-learn server.port=8080
```

SpringLearnApplicationTest.java

```
package com.cognizant.springlearn;
```

```

import org.junit.jupiter.api.Test;
import org.springframework.boot.test.context.SpringBootTest;
@SpringBootTest
class SpringLearnApplicationTests {

    @Test
    void contextLoads() {
    }

}

```

2) Spring Core – Load Country from Spring Configuration XML

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-bea
ns.xsd">  <bean id="country"
    class="com.cognizant.springlearn.Country">
    <property name="code" value="IN"/>

    <property name="name" value="India"/>
</bean>

```

</beans>

Country.java

```
package com.cognizant.springlearn;

public class Country {    private
String code;    private String name;

    public Country() {
        System.out.println("Inside Country Constructor.");
    }

    public String getCode() {
        System.out.println("Inside getCode()");
        return code;
    }

    public void setCode(String code) {
        System.out.println("Inside setCode()");
        this.code = code;
    }

    public String getName() {
        System.out.println("Inside getName()");
        return name;
    }

    public void setName(String name) {
        System.out.println("Inside setName()");
        this.name = name;
    }
}
```

```

@Override
public String toString() {
    return "Country [code=" + code + ", name=" + name + "];"
}
}

```

SpringLearnApplication.java

```

package com.cognizant.springlearn;

import org.springframework.context.ApplicationContext; import
org.springframework.context.support.ClassPathXmlApplicationContext
; public class SpringLearnApplication {    public static void main(String[]
args) {        displayCountry();
    }

    public static void displayCountry() {
ApplicationContext context =
        new ClassPathXmlApplicationContext("country.xml");

Country country =
        context.getBean("country", Country.class);
        System.out.println("Country : " + country);
    }
}

```

3) Hello World RESTful Web Service

HelloController.java

```

package com.cognizant.springlearn.controller;

import org.slf4j.Logger; import
org.slf4j.LoggerFactory;

import org.springframework.web.bind.annotation.GetMapping; import
org.springframework.web.bind.annotation.RestController;

@RestController public class
HelloController {    private static final
Logger LOGGER =

        LoggerFactory.getLogger(HelloController.class);

    @GetMapping("/hello")
public String sayHello() {
    LOGGER.info("START");

    String message = "Hello World!!";

LOGGER.info("END");    return
message;
    }
}

```

application.properties server.port=8083

spring.application.name=spring-learn **SpringLearnApplication.java**

```

package com.cognizant.springlearn;

import org.springframework.boot.SpringApplication;

import org.springframework.boot.autoconfigure.SpringBootApplication;

```

```
@SpringBootApplication public class
SpringLearnApplication {    public static
void main(String[] args) {
    SpringApplication.run(SpringLearnApplication.class, args);
}
}
```

pom.xml Dependency

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
</dependency>
```

4) REST - Country Web Service

Country.java

```
package com.cognizant.springlearn;

public class Country {    private String
code;    private String name;    public
Country() {
    }

    public String getCode() {
return code;
    }
}
```



```

    public void setCode(String code) {
this.code = code;
    }

    public String getName() {
return name;
    }

    public void setName(String name) {
this.name = name;
    }
}

```

country.xml

```

<?xml          version="1.0"          encoding="UTF-8"?>          <beans
xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">
    <bean id="country"
        class="com.cognizant.springlearn.Country">
    <property name="code" value="IN"/>
        <property name="name" value="India"/>
    </bean>
</beans>

```

CountryController.java

```
package com.cognizant.springlearn.controller; import
com.cognizant.springlearn.Country; import
org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping; import
org.springframework.web.bind.annotation.RestController;
@RestController
public class CountryController {
    @RequestMapping("/country")    public
    Country getCountryIndia() {
        ApplicationContext context =

            new ClassPathXmlApplicationContext("country.xml");
        Country country =

            context.getBean("country", Country.class);
        return country;
    }
}
```

5) REST - Get country based on country code

Country.java

```
package com.cognizant.springlearn;
public class Country {    private
String code;    private String name;
    public Country() {
```

```

    }

    public String getCode() {
return code;
    }

    public void setCode(String code) {
this.code = code;
    }

    public String getName() {
return name;
    }


    public void setName(String name) {
this.name = name;
    }
}

```

country.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/springbeans.xsd">
    <bean id="countryList"
class="java.util.ArrayList">

```

```
<constructor-arg>
  <list>
    <bean class="com.cognizant.springlearn.Country">
      <property name="code" value="US"/>
      <property name="name" value="United States"/>
    </bean>
    <bean class="com.cognizant.springlearn.Country">
      <property name="code" value="DE"/>
      <property name="name" value="Germany"/>
    </bean>

    <bean class="com.cognizant.springlearn.Country">
      <property name="code" value="IN"/>
      <property name="name" value="India"/>
    </bean>
    <bean class="com.cognizant.springlearn.Country">
      <property name="code" value="JP"/>
      <property name="name" value="Japan"/>
    </bean>
  </list>
</constructor-arg>
</bean>
</beans>
```

CountryService.java

```

package com.cognizant.springlearn.service; import
java.util.List;
import org.springframework.context.ApplicationContext; import
org.springframework.context.support.ClassPathXmlApplicationContext
; import org.springframework.stereotype.Service; import
com.cognizant.springlearn.Country;
@Service
public class CountryService {    public
Country getCountry(String code) {
ApplicationContext context =

        new ClassPathXmlApplicationContext("country.xml");
List<Country> countries =
        (List<Country>) context.getBean("countryList");
return countries.stream()
        .filter(c -> c.getCode().equalsIgnoreCase(code))
        .findFirst()
        .orElse(null);
    }
}

```

CountryController.java

```

package com.cognizant.springlearn.controller;
import
org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;

```

```

import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.Country; import
com.cognizant.springlearn.service.CountryService;
@RestController public class
CountryController {
    @Autowired
    private CountryService countryService;
    @GetMapping("/countries/{code}")
    public Country getCountry(
        @PathVariable String code) {    return
        countryService.getCountry(code);
    }
}

```

6) Create authentication service that returns JWT

AuthenticationResponse.java

```

package com.cognizant.springlearn.model;
public class AuthenticationResponse {    private
String token;
    public AuthenticationResponse(String token) {
this.token = token;
    }
}

```

```

    public String getToken() {
return token;
    }

    public void setToken(String token) {
this.token = token;
    }
}

```

AuthenticationController.java

```

package com.cognizant.springlearn.controller;

import java.nio.charset.StandardCharsets; import
java.util.Base64;
import org.springframework.web.bind.annotation.GetMapping; import
org.springframework.web.bind.annotation.RequestHeader; import
org.springframework.web.bind.annotation.RestController; import
com.cognizant.springlearn.model.AuthenticationResponse;
@RestController
public class AuthenticationController {
    @GetMapping("/authenticate")    public
AuthenticationResponse authenticate(
        @RequestHeader("Authorization") String authHeader) {
String base64Credentials =
        authHeader.substring("Basic ".length());
byte[] decodedBytes =

```

```

        Base64.getDecoder()
            .decode(base64Credentials);
String credentials =        new
String(decodedBytes,
        StandardCharsets.UTF_8);
String[] values =
credentials.split(":", 2);
String username = values[0];
String token =
    "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiI"
    + username
    + ".sampletoken";
return new AuthenticationResponse(token);
}
}

```

SecurityConfig.java

```

package com.cognizant.springlearn.config; import
org.springframework.context.annotation.Bean; import
org.springframework.context.annotation.Configuration; import
org.springframework.security.config.Customizer; import
org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.core.userdetails.User; import
org.springframework.security.core.userdetails.UserDetails; import

```



```

org.springframework.security.provisioning.InMemoryUserDetailsManager;
import org.springframework.security.web.SecurityFilterChain;

@Configuration public class
SecurityConfig {

    @Bean

    public InMemoryUserDetailsManager userDetailsService() {

        UserDetails user =

            User.withDefaultPasswordEncoder()

                .username("user")

                .password("pwd")

                .roles("USER")

                .build();

        return new InMemoryUserDetailsManager(user);
    }

    @Bean

    public SecurityFilterChain filterChain(
HttpSecurity http) throws Exception {        http

        .authorizeHttpRequests(auth ->
auth.anyRequest().authenticated())

        .httpBasic(Customizer.withDefaults());        return
http.build();

    }
}

```

pom.xml Dependency

```
<dependency>  
  <groupId>org.springframework.boot</groupId>  
  <artifactId>spring-boot-starter-security</artifactId>  
</dependency>
```